CHAPTER 13

GAS RESERVOIRS

Reservoirs containing only free gas are termed gas reservoirs. Such a reservoir contains a mixture of hydrocarbons, which exists wholly in the gaseous state. The mixture may be a *dry, wet,* or *condensate* gas, depending on the composition of the gas, along with the pressure and temperature at which the accumulation exists.

Gas reservoirs may have water influx from a contiguous water-bearing portion of the formation or may be volumetric (i.e., have no water influx).

Most gas engineering calculations involve the use of gas formation volume factor B_g and gas expansion factor E_g. Both factors are defined in Chapter 2 by Equations 2-52 through 2-56. Those equations are summarized below for convenience:

- Gas formation volume factor B_g is defined as the actual volume occupied by n moles of gas at a specified pressure and temperature, divided by the volume occupied by the same amount of gas at standard conditions. Applying the real gas equation-of-state to both conditions gives:

$$B_g = \frac{p_{sc}}{T_{sc}} \frac{zT}{p} = 0.02827 \frac{zT}{p} \qquad (13\text{-}1)$$

- The gas expansion factor is simply the reciprocal of B_g, or:

$$E_g = \frac{T_{sc}}{p_{sc}} \frac{p}{zT} = 35.37 \frac{p}{zT} \qquad (13\text{-}2)$$

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where B_g = gas formation volume factor, ft^3/scf
E_g = gas expansion factor, scf/ft^3

This chapter presents two approaches for estimating initial gas-in-place G, gas reserves, and the gas recovery for volumetric and water-drive mechanisms:

- Volumetric method
- Material balance approach

THE VOLUMETRIC METHOD

Data used to estimate the gas-bearing reservoir PV include, but are not limited to, well logs, core analyses, bottom-hole pressure (BHP), and fluid sample information, along with well tests. These data typically are used to develop various subsurface maps. Of these maps, structural and stratigraphic cross-sectional maps help to establish the reservoir's areal extent and to identify reservoir discontinuities, such as pinch-outs, faults, or gas-water contacts. Subsurface contour maps, usually drawn relative to a known or marker formation, are constructed with lines connecting points of equal elevation and therefore portray the geologic structure. Subsurface isopachous maps are constructed with lines of equal net gas-bearing formation thickness. With these maps, the reservoir PV can then be estimated by planimetering the areas between the isopachous lines and using an approximate volume calculation technique, such as the pyramidal or trapezoidal method.

The volumetric equation is useful in reserve work for estimating gas-in-place at any stage of depletion. During the development period before reservoir limits have been accurately defined, it is convenient to calculate gas-in-place per acre-foot of bulk reservoir rock. Multiplication of this unit figure by the best available estimate of bulk reservoir volume then gives gas-in-place for the lease, tract, or reservoir under consideration. Later in the life of the reservoir, when the reservoir volume is defined and performance data are available, volumetric calculations provide valuable checks on gas-in-place estimates obtained from material balance methods.

The equation for calculating gas-in-place is:

$$G = \frac{43,560\,Ah\phi(1 - S_{wi})}{B_{gi}} \tag{13-3}$$

where G = gas-in-place, scf
A = area of reservoir, acres
h = average reservoir thickness, ft
ϕ = porosity
S_{wi} = water saturation, and
B_{gi} = gas formation volume factor, ft^3/scf

This equation can be applied at both initial and abandonment conditions in order to calculate the recoverable gas.

Gas produced = Initial gas − Remaining gas

or

$$G_p = 43{,}560\,Ah\phi(1 - S_{wi})\left(\frac{1}{B_{gi}} - \frac{1}{B_{ga}}\right) \tag{13-4}$$

where B_{ga} is evaluated at abandonment pressure. Application of the volumetric method assumes that the pore volume occupied by gas is constant. If water influx is occurring, A, h, and S_w will change.

Example 13-1

A gas reservoir has the following characteristics:

$A = 3000$ acres $\quad h = 30$ ft $\quad \phi = 0.15 \quad S_{wi} = 20\%$
$T = 150°F \quad p_i = 2600$ psi

p	z
2600	0.82
1000	0.88
400	0.92

Calculate cumulative gas production and recovery factor at 1,000 and 400 psi.

Solution

Step 1. Calculate the reservoir pore volume P.V.

$$P.V = 43{,}560\ Ah\phi$$

$$P.V = 43{,}560\ (3000)\ (30)\ (0.15) = 588.06\ MMft^3$$

Step 2. Calculate B_g at every given pressure by using Equation 13-1.

p	z	B_g, ft^3/scf
2600	0.82	0.0054
1000	0.88	0.0152
400	0.92	0.0397

Step 3. Calculate initial gas-in-place at 2,600 psi.

$$G = 588.06\ (10^6)\ (1 - 0.2)/0.0054 = 87.12\ MMMscf$$

Step 4. Since the reservoir is assumed volumetric, calculate the remaining gas at 1,000 and 400 psi.

- Remaining gas at 1,000 psi

$$G_{1000\ psi} = 588.06(10^6)\ (1 - 0.2)/0.0152 = 30.95\ MMMscf$$

- Remaining gas at 400 psi

$$G_{400\ psi} = 588.06(10^6)\ (1 - 0.2)/0.0397 = 11.95\ MMMscf$$

Step 5. Calculate cumulative gas production G_p and the recovery factor RF at 1,000 and 400 psi.

- At 1,000 psi:

$$G_p = (87.12 - 30.95) \times 10^9 = 56.17\ MMM\ scf$$

$$RF = \frac{56.17 \times 10^9}{87.12 \times 10^9} = 64.5\%$$

- At 400 psi:

$$G_p = (87.12 - 11.95) \times 10^9 = 75.17 \text{ MMMscf}$$

$$RF = \frac{75.17 \times 10^9}{87.12 \times 10^9} = 86.3\%$$

The recovery factors for volumetric gas reservoirs will range from 80% to 90%. If a strong water drive is present, trapping of residual gas at higher pressures can reduce the recovery factor substantially, to the range of 50% to 80%.

THE MATERIAL BALANCE METHOD

If enough production-pressure history is available for a gas reservoir, the initial gas-in-place G, the initial reservoir pressure p_i, and the gas reserves can be calculated without knowing A, h, ϕ, or S_w. This is accomplished by forming a mass or mole balance on the gas as:

$$n_p = n_i - n_f \tag{13-5}$$

where n_p = moles of gas produced
n_i = moles of gas initially in the reservoir
n_f = moles of gas remaining in the reservoir

Representing the gas reservoir by an idealized gas container, as shown schematically in Figure 13-1, the gas moles in Equation 13-5 can be replaced by their equivalents using the real gas law to give:

$$\frac{p_{sc} G_p}{R\,T_{sc}} = \frac{p_i V}{z_i R T} - \frac{p[V - (W_e - W_p)]}{zRT} \tag{13-6}$$

where p_i = initial reservoir pressure
G_p = cumulative gas production, scf
p = current reservoir pressure
V = original gas volume, ft^3
z_i = gas deviation factor at p_i
z = gas deviation factor at p
T = temperature, °R
W_e = cumulative water influx, ft^3
W_p = cumulative water production, ft^3

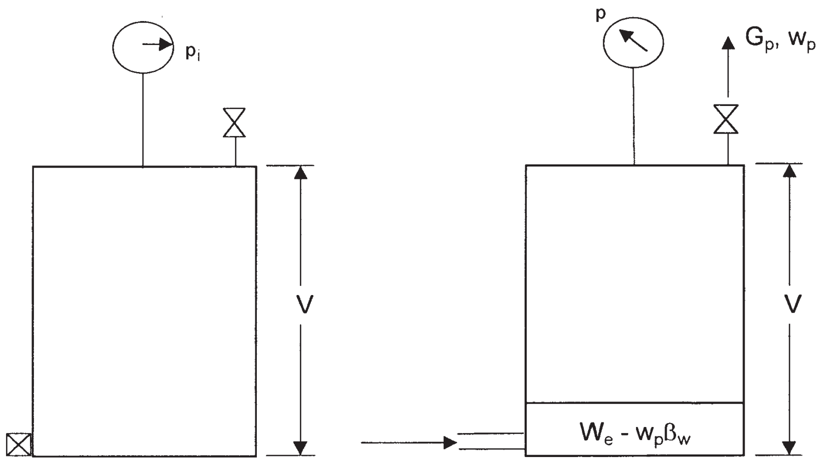

Figure 13-1. Idealized water-drive gas reservoir.

Equation 13-6 is essentially the general material balance equation (MBE). Equation 13-6 can be expressed in numerous forms depending on the type of the application and the driving mechanism. In general, dry gas reservoirs can be classified into two categories:

- Volumetric gas reservoirs
- Water-drive gas reservoirs

The remainder of this chapter is intended to provide the basic background in natural gas engineering. There are several excellent textbooks that comprehensively address this subject, including the following:

- Ikoku, C., *Natural Gas Reservoir Engineering,* 1984
- Lee, J. and Wattenbarger, R., *Gas Reservoir Engineering,* SPE, 1996

Volumetric Gas Reservoirs

For a volumetric reservoir and assuming no water production, Equation 13-6 is reduced to:

$$\frac{p_{sc} G_p}{T_{sc}} = \left(\frac{p_i}{z_i T}\right) V - \left(\frac{p}{zT}\right) V \tag{13-7}$$

Equation 13-7 is commonly expressed in the following two forms:

Form 1. In terms of p/z

Rearranging Equation 13-7 and solving for p/z gives:

$$\frac{p}{z} = \frac{p_i}{z_i} - \left(\frac{p_{sc}T}{T_{sc}V}\right)G_p \tag{13-8}$$

Equation 13-8 is an equation of a straight line when (p/z) is plotted versus the cumulative gas production G_p, as shown in Figure 13-2. This straight-line relationship is perhaps one of the most widely used relationships in gas-reserve determination.

The straight-line relationship provides the engineer with the reservoir characteristics:

• Slope of the straight line is equal to:

$$\text{slope} = \frac{p_{sc}T}{T_{sc}V} \tag{13-9}$$

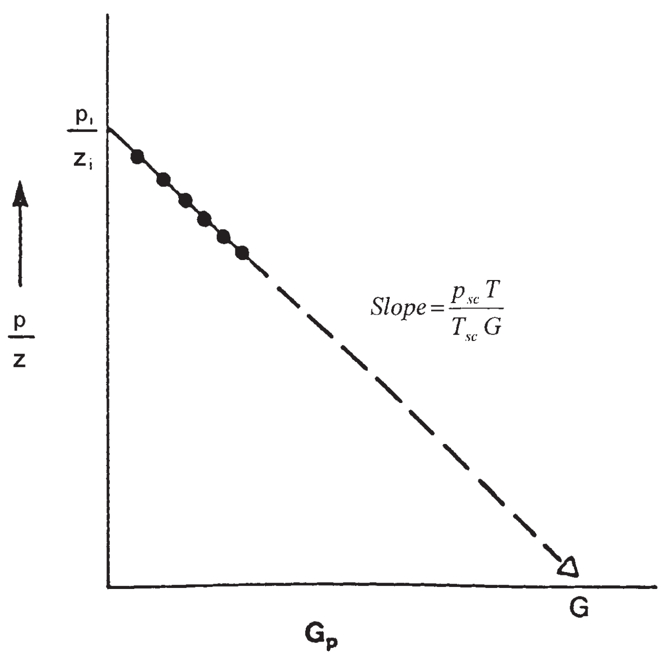

Figure 13-2. Gas material balance equation.

The original gas volume V can be calculated from the slope and used to determine the areal extent of the reservoir from:

$$V = 43{,}560\, Ah\, \phi\, (1 - S_{wi}) \tag{13-10}$$

where A is the reservoir area in acres.

- Intercept at $G_p = 0$ gives p_i/z_i
- Intercept at $p/z = 0$ gives the gas initially in place G in scf
- Cumulative gas production or gas recovery at any pressure

Example 13-2[1]

A volumetric gas reservoir has the following production history.

Time, t years	Reservoir Pressure, p psia	z	Cumulative Production, G_p MMMscf
0.0	1798	0.869	0.00
0.5	1680	0.870	0.96
1.0	1540	0.880	2.12
1.5	1428	0.890	3.21
2.0	1335	0.900	3.92

The following data are also available:

$\phi = 13\%$
$S_{wi} = 0.52$
$A = 1060$ acres
$h = 54$ ft
$T = 164°F$

Calculate the gas initially in place volumetrically and from the MBE.

Solution

Step 1. Calculate B_{gi} from Equation 13-1.

$$B_{gi} = 0.02827 \frac{(0.869)\,(164 + 460)}{1798} = 0.00853\ \text{ft}^3/\text{scf}$$

[1]After Ikoku, C., *Natural Gas Reservoir Engineering*, John Wiley & Sons, 1984.

Step 2. Calculate the gas initially in place volumetrically by applying Equation 13-3.

$$G = 43{,}560\ (1060)\ (54)\ (0.13)\ (1 - 0.52)/0.00853 = 18.2\ \text{MMMscf}$$

Step 3. Plot p/z versus G_p as shown in Figure 13-3 and determine G.

$$G = 14.2\ \text{MMMscf}$$

This checks the volumetric calculations.

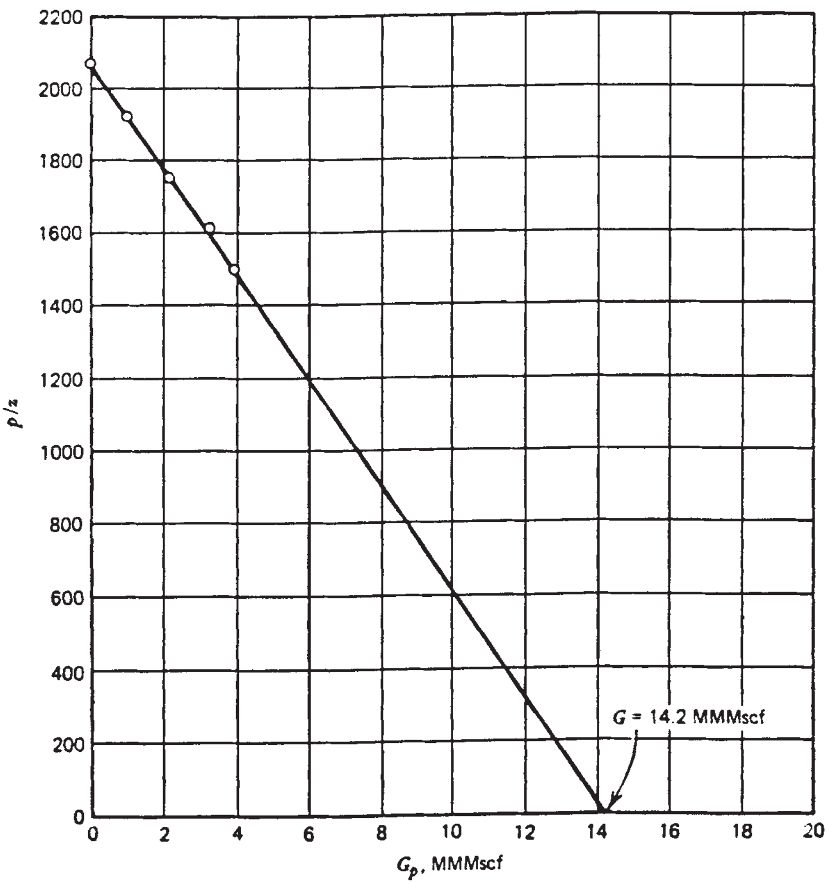

Figure 13-3. Relationship of p/z vs. G_p for Example 13-2.

The initial reservoir gas volume V can be expressed in terms of the volume of gas at standard conditions by:

$$V = B_g\,G = \left(\frac{p_{sc}}{T_{sc}}\frac{z_i\,T}{p_i}\right)G$$

Combining the above relationship with that of Equation 13-8 gives:

$$\frac{p}{z} = \frac{p_i}{z_i} - \left[\left(\frac{p_i}{z_i}\right)\frac{1}{G}\right]G_p \qquad (13\text{-}11)$$

This relationship can be expressed in a more simplified form as:

$$\frac{p}{Z} = \frac{p_i}{Z_i} - \left[m\right]G_p$$

where the coefficient m is essentially constant and represents the resulting straight line when P/Z is plotted against G_P. The slope, m is defined by:

$$m = \left(\frac{p_i}{Z_i}\right)\frac{1}{G}$$

Equivalently, m is defined by Equation 13-9 as:

$$m = \frac{T\,p_{sc}}{T_{sc}\,V}$$

where
G = Original gas-in-place, scf
V = Original gas-in-place, ft^3

Again, Equation 13-11 shows that for a volumetric reservoir, the relationship between (p/z) and G_p is essentially linear. This popular equation indicates that by extrapolation of the straight line to abscissa, i.e., at $p/z = 0$, will give the value of the gas initially in place as $G = G_p$.

The graphical representation of Equation 13-11 can be used to detect the presence of water influx, as shown graphically in Figure 13-4. When the plot of (p/z) versus G_p deviates from the linear relationship, it indicates the presence of water encroachment.

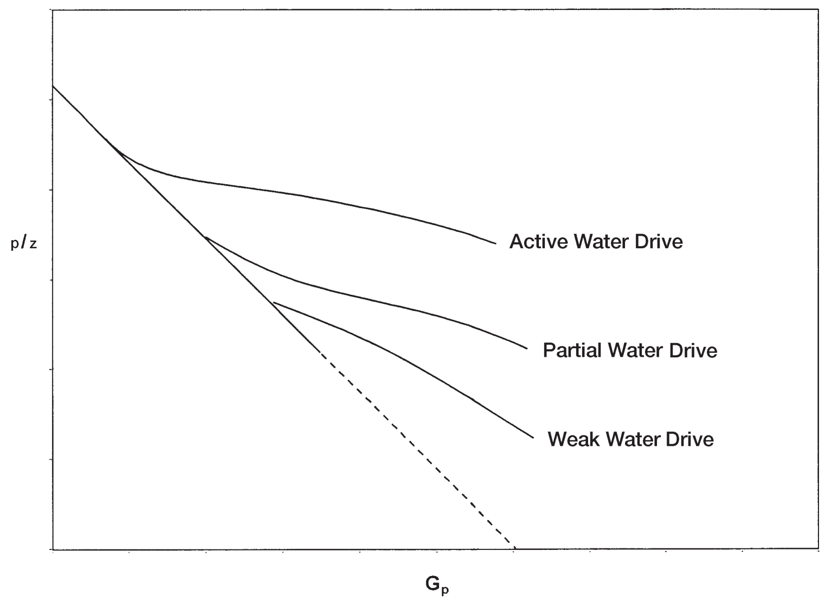

Figure 13-4. Effect of water drive on p/z vs. G_p relationship.

Many other graphical methods have been proposed for solving the gas MBE that are useful in detecting the presence of water influx. One such graphical technique is called the **energy plot,** which is based on arranging Equation 13-11 and taking the logarithm of both sides to give:

$$\log\left[1-\frac{z_i\,p}{p_i\,z}\right]=\log G_p-\log G \tag{13-12}$$

Figure 13-5 shows a schematic illustration of the plot.

From Equation 13-12, it is obvious that a plot of $[1 - (z_i\ p)/(p_i\ z)]$ versus G_p on log-log coordinates will yield a straight line with a slope of one (45° angle). An extrapolation to one on the vertical axis ($p = 0$) yields a value for initial gas-in-place, G. The graphs obtained from this type of analysis have been referred to as *energy plots.* They have been found to be useful in detecting water influx early in the life of a reservoir. If W_e is not zero, the slope of the plot will be less than one, and will also decrease with time, since W_e increases with time. An increasing slope can only occur as a result of either gas leaking from the reservoir or bad data,

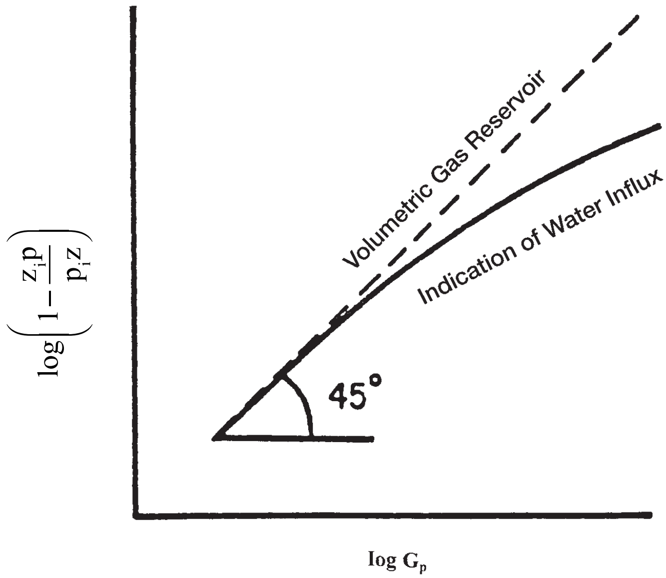

Figure 13-5. An energy plot.

since the increasing slope would imply that the gas-occupied pore volume was increasing with time.

It should be pointed out that the average field, $(p/Z)_{Field}$, can be estimated from the individual wells' p/Z versus G_P performance by applying the following relationship:

$$\left(\frac{p}{Z}\right)_{Field} = \frac{p_i}{Z_i} - \frac{\sum_{j=1}^{n}(G_P)_j}{\sum_{j=1}^{n}\left[\dfrac{G_P}{\dfrac{p_i}{Z_i} - \dfrac{p}{Z}}\right]_j}$$

The summation Σ is taking over the total number n of the field gas wells, that is, j = 1, 2, ... n. The total field performance in terms of $(p/Z)_{Field}$ versus $(G_P)_{Field}$ can then be constructed from the estimated values of the field p/Z and actual total field production, that is, $(p/Z)_{Field}$ versus ΣG_P. The above equation is applicable as long as all wells are producing with defined static boundaries, that is, under pseudosteady-state conditions.

When using the MBE for reserve analysis for an entire reservoir that is characterized by a distinct lack of pressure equilibrium throughout, the following average reservoir pressure decline, $(p/Z)_{Field}$, can be used:

$$\left(\frac{p}{Z}\right)_{Field} = \frac{\sum_{j=1}^{n}\left(\frac{p\,\Delta G_P}{\Delta p}\right)_j}{\sum_{j=1}^{n}\left(\frac{\Delta G_P}{\Delta p / Z}\right)_j}$$

where Δp and ΔG_P are the incremental pressure difference and cumulative production, respectively.

The gas recovery factor (RF) at any depletion pressure is defined as the cumulative gas produced, G_P, at this pressure divided by the gas initially in place, G:

$$RF = \frac{G_P}{G}$$

Introducing the gas RF to Equation 8-60 gives

$$\frac{p}{Z} = \frac{p_i}{Z_i}\left[1 - \frac{G_P}{G}\right]$$

or

$$\frac{p}{Z} = \frac{p_i}{Z_i}\left[1 - RF\right]$$

Solving for the recovery factor at any depletion pressure gives:

$$RF = 1 - \left[\frac{Z_i}{Z}\,\frac{p}{p_i}\right]$$

Form 2. In terms of B_g

From the definition of the gas formation volume factor, it can be expressed as:

$$B_{gi} = \frac{V}{G}$$

Combining the above expression with Equation 13-1 gives:

$$\frac{p_{sc}}{T_{sc}} \frac{z_i T}{p_i} = \frac{V}{G} \tag{13-13}$$

where V = volume of gas originally in place, ft^3
G = volume of gas originally in place, scf
p_i = original reservoir pressure
z_i = gas compressibility factor at p_i

Equation 13-13 can be combined with Equation 13-7, to give:

$$G = \frac{G_p B_g}{B_g - B_{gi}} \tag{13-14}$$

Equation 13-14 suggests that to calculate the initial gas volume, the only information required is production data, pressure data, gas specific gravity for obtaining z-factors, and reservoir temperature. Early in the producing life of a reservoir, however, the denominator of the right-hand side of the material balance equation is very small, while the numerator is relatively large. A small change in the denominator will result in a large discrepancy in the calculated value of initial gas-in-place. Therefore, the material balance equation should not be relied on early in the producing life of the reservoir.

Material balances on volumetric gas reservoirs are simple. Initial gas-in-place may be computed from Equation 13-14 by substituting cumulative gas produced and appropriate gas formation volume factors at corresponding reservoir pressures during the history period. If successive calculations at various times during the history give consistent values for initial gas-in-place, the reservoir is operating under volumetric control and computed G is reliable, as shown in Figure 13-6. Once G has been determined and the absence of water influx established in this fashion,

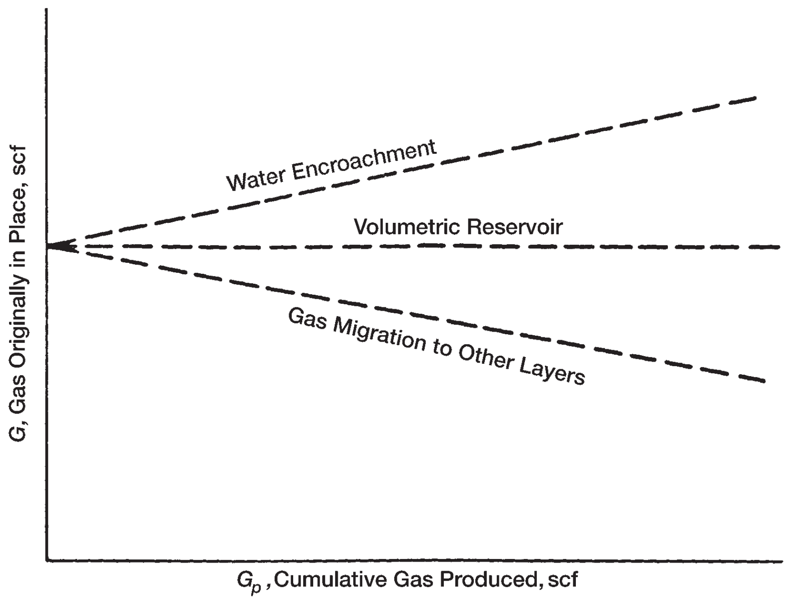

Figure 13-6. Graphical determination of the gas initially in place G.

the same equation can be used to make future predictions of cumulative gas production function of reservoir pressure.

Ikoku (1984) points out that successive application of Equation 13-14 will normally result in increasing values of the gas initially in place G with time if water influx is occurring. If there is gas leakage to another zone due to bad cement jobs or casing leaks, however, the computed value of G may decrease with time.

Example 13-3

After producing 360 MMscf of gas from a volumetric gas reservoir, the pressure has declined from 3,200 psi to 3,000 psi, given:

$B_{gi} = 0.005278\ ft^3/scf$
$B_g = 0.005390\ ft^3/scf$

a. Calculate the gas initially in place.
b. Recalculate the gas initially in place assuming that the pressure measurements were incorrect and the true average pressure is 2,900 psi. The gas formation volume factor at this pressure is $0.00558\ ft^3/scf$.

Solution

a. Using Equation 13-14, calculate G.

$$G = \frac{360 \times 10^6 (0.00539)}{0.00539 - 0.005278} = 17.325 \text{ MMMscf}$$

b. Recalculate G by using the correct value of B_g.

$$G = \frac{360 \times 10^6 (0.00668)}{0.00558 - 0.005278} = 6.652 \text{ MMMscf}$$

Thus, an error of 100 psia, which is only 3.5% of the total reservoir pressure, resulted in an increase in calculated gas-in-place of approximately 160%, a 2½-fold increase. Note that a similar error in reservoir pressure later in the producing life of the reservoir will not result in an error as large as that calculated early in the producing life of the reservoir.

Water-Drive Gas Reservoirs

If the gas reservoir has a water drive, then there will be two unknowns in the material balance equation, even though production data, pressure, temperature, and gas gravity are known. These two unknowns are initial gas-in-place and cumulative water influx. In order to use the material balance equation to calculate initial gas-in-place, some independent method of estimating W_e, the cumulative water influx, must be developed as discussed in Chapter 11.

Equation 13-14 can be modified to include the cumulative water influx and water production to give:

$$G = \frac{G_p B_g - (W_e - W_p B_w)}{B_g - B_{gi}} \tag{13-15}$$

The above equation can be arranged and expressed as:

$$G + \frac{W_e}{B_g - B_{gi}} = \frac{G_p B_g + W_p B_w}{B_g - B_{gi}} \tag{13-16}$$

Equation 13-16 reveals that for a volumetric reservoir, i.e., $W_e = 0$, the right-hand side of the equation will be constant regardless of the amount of gas G_p that has been produced. For a water-drive reservoir, the values of the right-hand side of Equation 13-16 will continue to increase because of the $W_e/(B_g - B_{gi})$ term. A plot of several of these values at successive time intervals is illustrated in Figure 13-7. Extrapolation of the line formed by these points back to the point where $G_p = 0$ shows the true value of G, because when $G_p = 0$, then $W_e/(B_g - B_{gi})$ is also zero.

This graphical technique can be used to estimate the value of W_e, because at any time the difference between the horizontal line (i.e., true value of G) and the sloping line $[G + (W_e)/(B_g - B_{gi})$ will give the value of $W_e/(B_g - B_{gi})$.

Because gas often is bypassed and trapped by the encroaching water, recovery factors for gas reservoirs with water drive can be significantly lower than for volumetric reservoirs produced by simple gas expansion. In addition, the presence of reservoir heterogeneities, such as low-permeability stringers or layering, may reduce gas recovery further. As noted previously, ultimate recoveries of 80% to 90% are common in volumetric gas reservoirs, while typical recovery factors in water-drive gas reservoirs can range from 50% to 70%.

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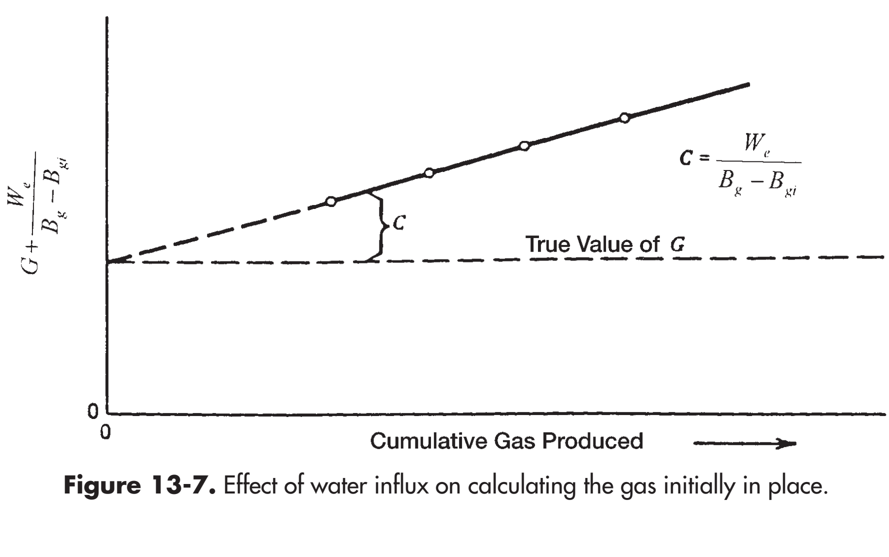

Figure 13-7. Effect of water influx on calculating the gas initially in place.

ultimate recoveries of 80% to 90% are common in volumetric gas reservoirs, while typical recovery factors in water-drive gas reservoirs can range from 50% to 70%. The amount of gas that is trapped in a region that has been flooded by water encroachment can be estimated by defining the following characteristic reservoir parameters and taking the steps outlined below:

(P.V) = reservoir pore volume, ft^3
$(P.V)_{water}$ = pore volume of the water-invaded zone, ft^3
S_{grw} = residual gas saturation to water displacement
S_{wi} = initial water saturation
G = gas initially in place, scf
G_P = cumulative gas production at depletion pressure p, scf
B_{gi} = initial gas formation volume factor, ft^3/scf
B_g = gas formation volume factor at depletion pressure p, ft^3/scf
Z = gas deviation factor at depletion pressure p

Step 1. Express the reservoir pore volume, (P.V), in terms of the initial gas-in-place, G, as follows:

$$G\, B_{gi} = (P.V)\,(1 - S_{wi})$$

Solving for the reservoir pore volume gives:

$$(P.V) = \frac{G B_{gi}}{1 - S_{wi}}$$

Step 2. Calculate the pore volume in the water-invaded zone:

$$W_e - W_p\, B_w = (P.V)_{water}\,(1 - S_{wi} - S_{grw})$$

Solving for the pore volume of the water-invaded zone, $(P.V)_{water}$, gives:

$$(P.V)_{water} = \frac{W_e - W_p B_w}{1 - S_{wi} - S_{grw}}$$

Step 3. Calculate trapped gas volume in the water-invaded zone, or:

$$\text{Trapped gas volume} = (P.V)_{water}\, S_{grw}$$

$$\text{Trapped gas volume} = \left[\frac{W_e - W_p\, B_w}{1 - S_{wi} - S_{grw}}\right] S_{grw}$$

Step 4. Calculate the number n of moles of gas trapped in the water-invaded zone by using the equation of state, or:

$$p\,(\text{Trapped gas volume}) = Z\, n\, R\, T$$

Solving for n gives:

$$n = \frac{p\left[\dfrac{W_e - W_p\, B_w}{1 - S_{wi} - S_{grw}}\right] S_{grw}}{Z\, R\, T}$$

which indicates that the higher the pressure, the greater the quantity of trapped gas. Dake (1994) points out that if the pressure is reduced by rapid gas withdrawal, the volume of gas trapped in each individual pore space, that is, S_{grw}, will remain unaltered, but its quantity, n, will be reduced.

Step 5. The gas saturation at any pressure can be adjusted to account for the trapped gas, as follows:

$$S_g = \frac{\text{remaining gas volume} - \text{trapped gas volume}}{\text{reservoir pore volume} - \text{pore volume of water invaded zone}}$$

$$S_g = \frac{\left(G - G_p\right) B_g - \left[\dfrac{W_e - W_p\, B_w}{1 - S_{wi} - S_{grw}}\right] S_{grw}}{\left(\dfrac{G\, B_{gi}}{1 - S_{wi}}\right) - \left[\dfrac{W_e - W_p\, B_w}{1 - S_{wi} - S_{grw}}\right]}$$

MATERIAL BALANCE EQUATION AS A STRAIGHT LINE

Havlena and Odeh (1963) expressed the material balance in terms of gas production, fluid expansion, and water influx as:

$$\begin{matrix}\text{Underground} \\ \text{withdrawal}\end{matrix} = \begin{matrix}\text{Gas} \\ \text{expansion}\end{matrix} + \begin{matrix}\text{Water expansion/} \\ \text{pore compaction}\end{matrix} + \begin{matrix}\text{Water} \\ \text{influx}\end{matrix}$$

or

$$G_p B_g + W_p B_w = G(B_g - B_{gi}) + G B_{gi} \frac{(c_w S_{wi} + c_f)}{1 - S_{wi}} \Delta p + W_e B_w \tag{13-17}$$

Using the nomenclature of Havlena and Odeh, as described in Chapter 11, gives:

$$F = G\,(E_g + E_{f,w}) + W_e\, B_w \tag{13-18}$$

with the terms F, E_g, and $E_{f,w}$ as defined by:

- Underground fluid withdrawal F:

$$F = G_p\, B_g + W_p\, B_w \tag{13-19}$$

- Gas expansion E_g:

$$E_g = B_g - B_{gi} \tag{13-20}$$

- Water and rock expansion $E_{f,w}$:

$$E_{f,w} = B_{gi} \frac{(c_w S_{wi} + c_f)}{1 - S_{wi}} \tag{13-21}$$

Assuming that the rock and water expansion term $E_{f,w}$ is negligible in comparison with the gas expansion E_g, Equation 13-18 is reduced to:

$$F = G\, E_g + W_e\, B_w \tag{13-22}$$

Finally, dividing both sides of the equation by E_g gives:

$$\frac{F}{E_g} = G + \frac{W_e B_w}{E_g} \tag{13-23}$$

Using the production, pressure, and PVT data, the left-hand side of this expression should be plotted as a function of the cumulative gas production, G_p. This is simply for display purposes to inspect its variation during depletion. Plotting F/E_g versus production time or pressure decline, Δp, can be equally illustrative.

Dake (1994) presented an excellent discussion of the strengths and weaknesses of the MBE as a straight line. He points out that the plot will have one of the three shapes depicted in Figure 13-8. If the reservoir is of the volumetric depletion type, $W_e = 0$, then the values of F/E_g evaluated, say, at six monthly intervals, should plot as a straight line parallel to the abscissa—whose ordinate value is the GIIP.

Alternatively, if the reservoir is affected by natural water influx, then the plot of F/E_g will usually produce a concave downward shaped arc

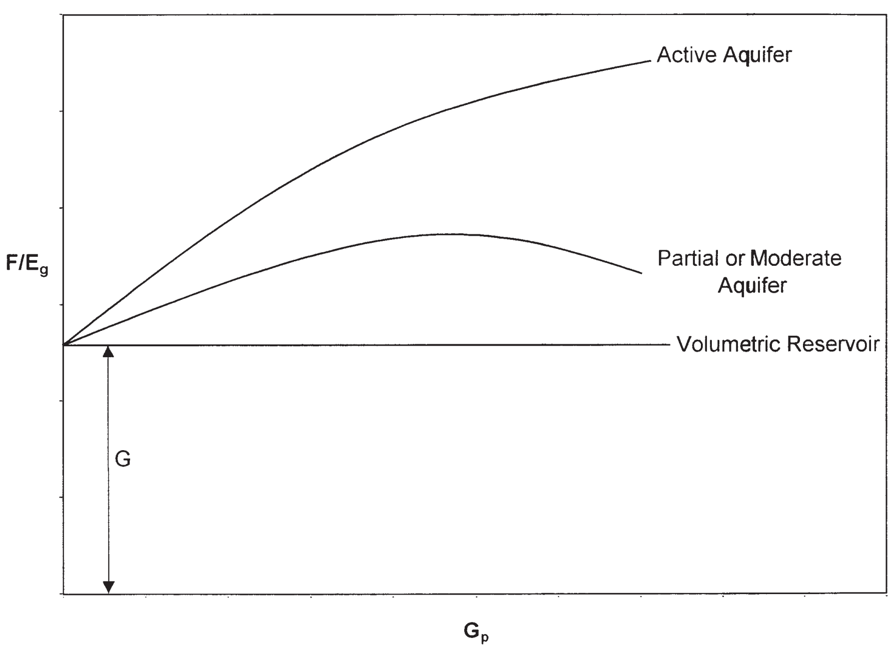

Figure 13-8. Defining the reservoir-driving mechanism.

whose exact form is dependent upon the aquifer size and strength and the gas off-take rate. Backward extrapolation of the F/E_g trend to the ordinate should nevertheless provide an estimate of the GIIP ($W_e \sim 0$); however, the plot can be highly nonlinear in this region yielding a rather uncertain result. The main advantage in the F/E_g versus G_p plot is that it is much more sensitive than other methods in establishing whether the reservoir is being influenced by natural water influx or not.

The graphical presentation of Equation 13-23 is illustrated by Figure 13-9. A graph of F/E_g vs. $\Sigma\Delta p\ W_{eD}/E_g$ yields a straight line, provided the unsteady-state influx summation, $\Sigma\Delta p\ W_{eD}$, is accurately assumed. The resulting straight line intersects the y-axis at the initial gas-in-place G and has a slope equal to the water influx constant B.

Nonlinear plots will result if the aquifer is improperly characterized. A systematic upward or downward curvature suggests that the summation term is too small or too large, respectively, while an S-shaped curve indicates that a linear (instead of a radial) aquifer should be assumed. The points should plot sequentially from left to right. A reversal of this

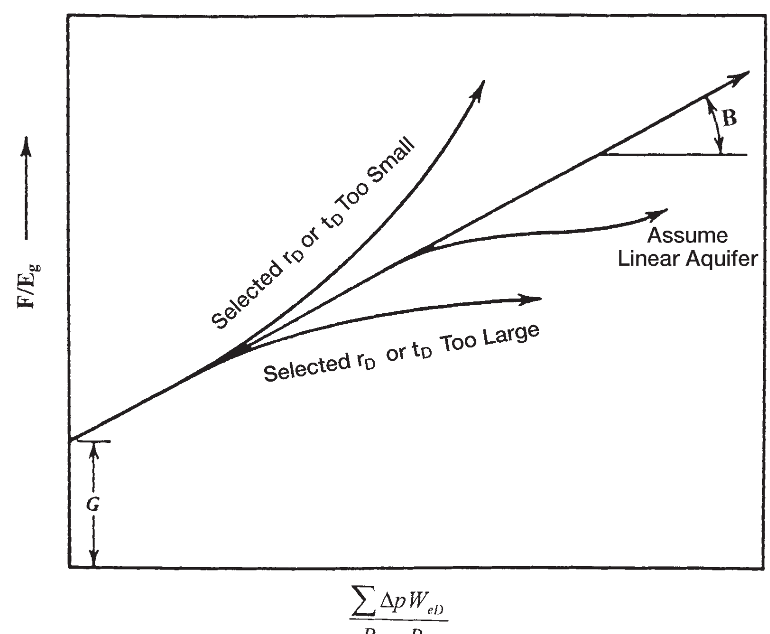

Figure 13-9. Havlena-Odeh MBE plot for a gas reservoir.

plotting sequence indicates that an unaccounted aquifer boundary has been reached and that a smaller aquifer should be assumed in computing the water influx term.

A linear infinite system rather than a radial system might better represent some reservoirs, such as reservoirs formed as fault blocks in salt domes. The van Everdingen-Hurst dimensionless water influx W_{eD} is replaced by the square root of time as:

$$W_e = C \sum \Delta p_n \sqrt{t - t_n} \tag{13-24}$$

where C = water influx constant ft^3/psi
t = time (any convenient units, i.e., days, year)

The water influx constant C must be determined by using the past production and pressure of the field in conjunction with Havlena-Odeh methodology. For the linear system, the underground withdrawal F is plotted versus $[\Sigma\, \Delta p_n\, zt - t_n/(B - B_{gi})]$ on a Cartesian coordinate graph. The plot should result in a straight line with G being the intercept and the water influx constant C being the slope of the straight line.

To illustrate the use of the linear aquifer model in the gas MBE as expressed as an equation of straight line, i.e., Equation 13-23, Havlena and Odeh proposed the following problem.

Example 13-4

The volumetric estimate of the gas initially in place for a dry-gas reservoir ranges from 1.3 to 1.65×10^{12} scf. Production, pressures, and pertinent gas expansion term, i.e., $E_g = B_g - B_{gi}$, are presented in Table 13-1. Calculate the original gas-in-place G.

Solution

Step 1. Assume volumetric gas reservoir.

Step 2. Plot (p/z) versus G_p or $G_p\, B_g/(B_g - B_{gi})$ versus G_p.

Step 3. A plot of $G_p\, B_g/(B_g - B_{gi})$ vs. $G_p\, B_g$ showed an upward curvature, as shown in Figure 13-10, indicating water influx.

Table 13-1
Havlena-Odeh Dry-Gas Reservoir Data for Example 13-4

Time (months)	Average Reservoir Pressure (psi)	$E_g = (B_g - B_{gi}) \times 10^{-6}$ (ft³/scf)	$F = (G_b B_g) \times 10^6$ (ft³)	$\frac{\Sigma \Delta p_n\ zt - t_n}{B_g - B_{gi}}$ (10^6)	$F/E_g = \frac{G_p B_g}{B_g - B_{gi}}$ (10^{12})
0	2883	0.0	—	—	—
2	2881	4.0	5.5340	0.3536	1.3835
4	2874	18.0	24.5967	0.4647	1.3665
6	2866	34.0	51.1776	0.6487	1.5052
8	2857	52.0	76.9246	0.7860	1.4793
10	2849	68.0	103.3184	0.9306	1.5194
12	2841	85.0	131.5371	1.0358	1.5475
14	2826	116.5	180.0178	1.0315	1.5452
16	2808	154.5	240.7764	1.0594	1.5584
18	2794	185.5	291.3014	1.1485	1.5703
20	2782	212.0	336.6281	1.2426	1.5879
22	2767	246.0	392.8592	1.2905	1.5970
24	2755	273.5	441.3134	1.3702	1.6136
26	2741	305.5	497.2907	1.4219	1.6278
28	2726	340.0	556.1110	1.4672	1.6356
30	2712	373.5	613.6513	1.5174	1.6430
32	2699	405.0	672.5969	1.5714	1.6607
34	2688	432.5	723.0868	1.6332	1.6719
36	2667	455.5	771.4902	1.7016	1.6937

Step 4. Assuming a linear water influx, plot $G_p\ B_g/(B_g - B_{gi})$ versus $\left[\sum \Delta p_n \sqrt{t - t_n}\right] / (B_g - B_{gi})$ as shown in Figure 13-11.

Step 5. As evident from Figure 13-11, the necessary straight-line relationship is regarded as satisfactory evidence of the presence of a linear aquifer.

Step 6. From Figure 13-11, determine the original gas-in-place G and the linear water influx constant C as:

$$G = 1.325 \times 10^{12} \text{ scf}$$

$$C = 212.7 \times 10^3 \text{ ft}^3/\text{psi}$$

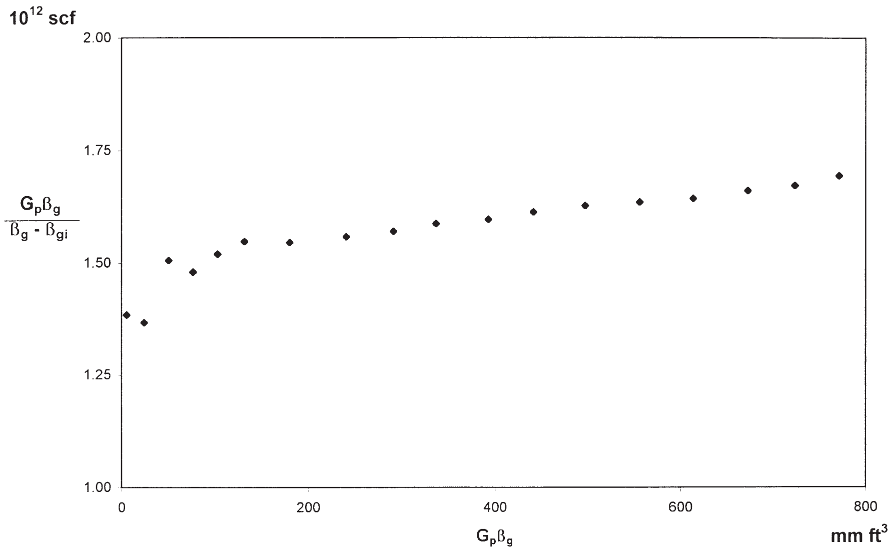

Figure 13-10. Indication of the water influx.

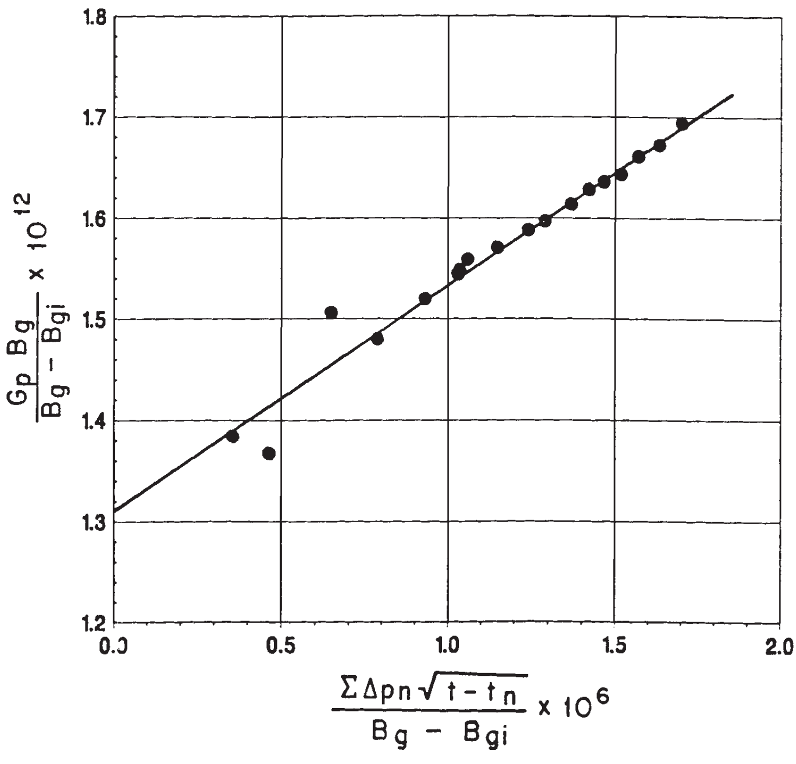

Figure 13-11. Havlena-Odeh MBE plot for Example 13-4.

ABNORMALLY PRESSURED GAS RESERVOIRS

Hammerlindl (1971) pointed out that in abnormally high-pressure volumetric gas reservoirs, two distinct slopes are evident when the plot of p/z versus G_p is used to predict reserves because of the formation and fluid compressibility effects as shown in Figure 13-12. The final slope of the p/z plot is steeper than the initial slope; consequently, reserve estimates based on the early life portion of the curve are erroneously high. The initial slope is due to gas expansion and significant pressure maintenance brought about by formation compaction, crystal expansion, and water expansion. At an approximately normal pressure gradient, the formation compaction is essentially complete and the reservoir assumes the characteristics of a normal gas expansion reservoir. This accounts for the second slope. Most early decisions are made based on the early life extrapolation of the p/z plot; therefore, the effects of hydrocarbon

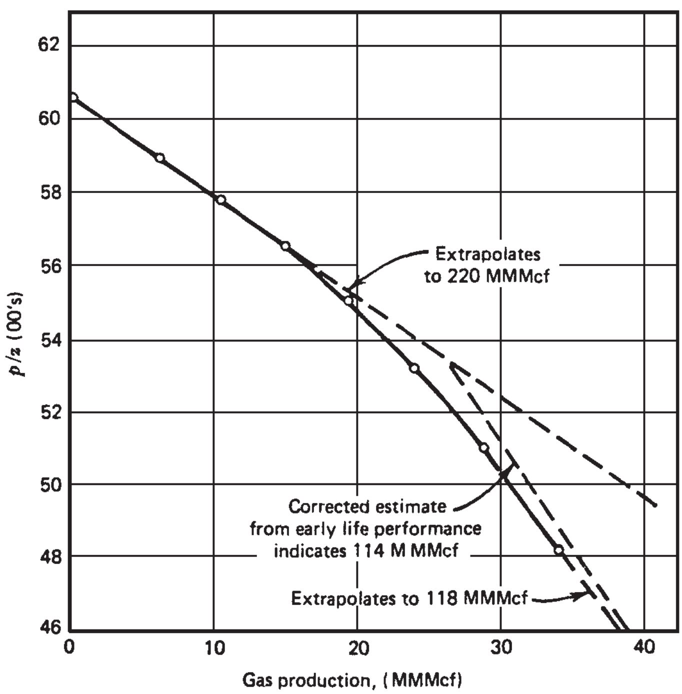

Figure 13-12. P/z versus cumulative production. North Ossum Field, Lafayette Parish, Louisiana NS2B Reservoir. (*After Hammerlindl.*)

pore volume change on reserve estimates, productivity, and abandonment pressure must be understood.

All gas reservoir performance is related to effective compressibility, not gas compressibility. When the pressure is abnormal and high, effective compressibility may equal two or more times that of gas compressibility. If effective compressibility is equal to twice the gas compressibility, then the first cubic foot of gas produced is due to 50% gas expansion and 50% formation compressibility and water expansion. As the pressure is lowered in the reservoir, the contribution due to gas expansion becomes greater because gas compressibility is approaching effective compressibility. Using formation compressibility, gas production, and shut-in bottom-hole pressures, two methods are presented for correcting the reserve estimates from the early life data (assuming no water influx).

Roach (1981) proposed a graphical technique for analyzing abnormally pressured gas reservoirs. The MBE as expressed by Equation 13-17 may be written in the following form for a volumetric gas reservoir:

$$(p/z)c_t = (p_i/z_i) - \left[1 - \frac{G_p}{G}\right] \tag{13-25}$$

where

$$c_t = 1 - \frac{(c_f + c_w S_{wi})(p_i - p)}{1 - S_{wi}} \tag{13-26}$$

Defining the rock expansion term E_R as:

$$E_R = \frac{c_f + c_w S_{wi}}{1 - S_{wi}} \tag{13-27}$$

Equation 13-26 can be expressed as:

$$c_t = 1 - E_R\,(p_i - p) \tag{13-28}$$

Equation 13-25 indicates that plotting $(p/z)c_t$ versus cumulative gas production on Cartesian coordinates results in a straight line with an x-intercept at the original gas-in-place and a y-intercept at the original p/z. Since c_t is unknown and must be found by choosing the compressibility values resulting in the best straight-line fit, this method is a trial-and-error procedure.

Roach used the data published by Duggan (1972) for the Mobil-David Anderson gas field to illustrate the application of Equations 13-25 and 13-28 to determine graphically the gas initially in place. Duggan reported that the reservoir had an initial pressure of 9,507 psig at 11,300 ft. Volumetric estimates of original gas-in-place indicated that the reservoir contains 69.5 MMMscf. The historical p/z versus G_p plot produced an initial gas-in-place of 87 MMMscf, as shown in Figure 13-13.

Using the trial-and-error approach, Roach showed that a value of the rock expansion term E_R of 18.5×10^{-6} would result in a straight line with a gas initially in place of 75 MMMscf, as shown in Figure 13-13.

To avoid the trial-and-error procedure, Roach proposed that Equations 13-25 and 13-28 can be combined and expressed in a linear form by:

$$\alpha = \left(\frac{1}{G}\right)\beta - E_R \tag{13-29}$$

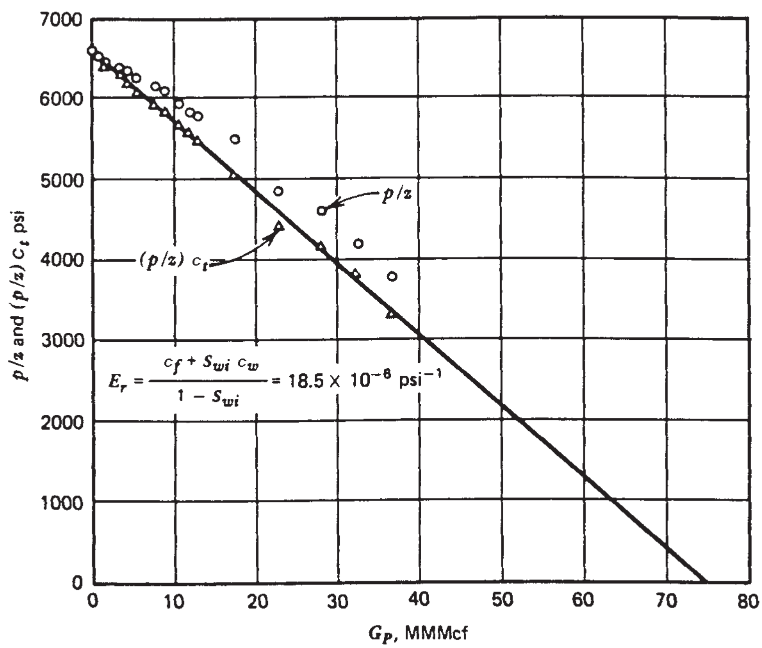

Figure 13-13. Mobil-David Anderson "L" p/z versus cumulative production. (*After Roach.*)

with

$$\alpha = \frac{[(p_i/z_i)/(p/z)] - 1}{(p_i - p)} \tag{13-30}$$

$$\beta = \frac{(p_i/z_i)\,(p/z)}{(p_i - p)} \tag{13-31}$$

where G = initial gas-in-place, scf
E_R = rock expansion term, psi^{-1}
S_{wi} = initial water saturation

Roach (1981) shows that a plot of α versus β will yield a straight line with slope 1/G and y-intercept = $-E_R$. To illustrate his proposed methodology, he applied Equation 13-29 to the Mobil-David gas field as shown in Figure 13-14. The slope of the straight line gives G = 75.2 MMMscf and the intercept gives $E_R = 18.5 \times 10^{-6}$.

Begland and Whitehead (1989) proposed a method to predict the percent recovery of volumetric, high-pressured gas reservoirs from the initial pressure to the abandonment pressure with only initial reservoir data. The proposed technique allows the pore volume and water compressibilities to be pressure-dependent. The authors derived the following form of the MBE for a volumetric gas reservoir:

$$r = \frac{G_p}{G} = \frac{B_g - B_{gi}}{B_g} + \frac{\dfrac{B_{gi}\,S_{wi}}{1 - S_{wi}}\left[\dfrac{B_{tw}}{B_{twi}} - 1 + \dfrac{c_f\,(p_i - p)}{S_{wi}}\right]}{B_g} \tag{13-32}$$

where r = recovery factor
B_g = gas formation volume factor, bbl/scf
c_f = formation compressibility, psi^{-1}
B_{tw} = two-phase water formation volume factor, bbl/STB
B_{twi} = initial two-phase water formation volume factor, bbl/STB

The water two-phase FVF is determined from:

$$B_{tw} = B_w + B_g\,(R_{swi} - R_{sw}) \tag{13-33}$$

where R_{sw} = gas solubility in the water phase, scf/STB
B_w = water FVF, bbl/STB

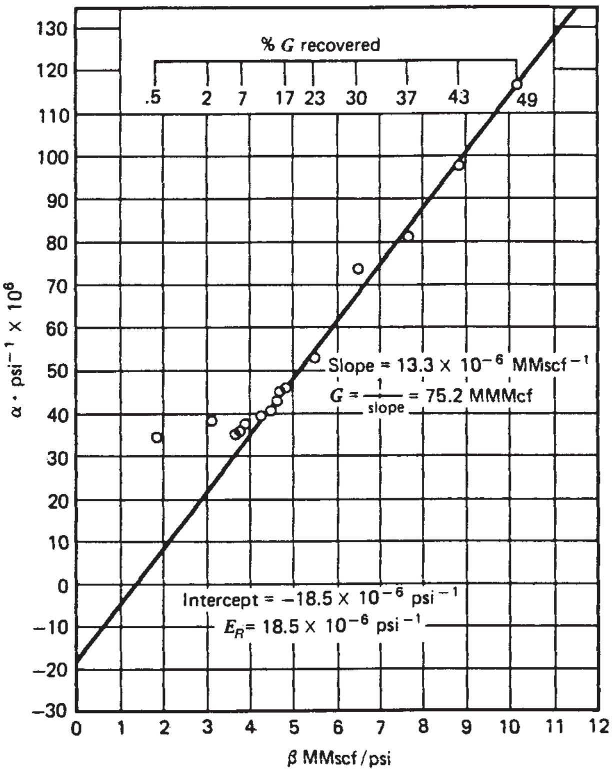

Figure 13-14. Mobil-David Anderson "L" gas material balance. (*After Roach.*)

The following three assumptions are inherent in Equation 13-32:

- A volumetric, single-phase gas reservoir
- No water production
- The formation compressibility c_f remains constant over the pressure drop $(p_i - p)$

The authors point out that the changes in water compressibility c_w are implicit in the change of B_{tw} with pressure as determined by Equation 13-33.

Begland and Whitehead suggest that because c_f is pressure-dependent, Equation 13-32 is not correct as reservoir pressure declines from the initial pressure to some value several hundred psi lower. The pressure dependence of c_f can be accounted for in Equation 13-32 and is solved in an incremental manner.

Effect of Gas Production Rate on Ultimate Recovery

Volumetric gas reservoirs are essentially depleted by expansion and, therefore, the ultimate gas recovery is independent of the field production rate. The gas saturation in this type of reservoir is never reduced; only the number of pounds of gas occupying the pore spaces is reduced. Therefore, it is important to reduce the abandonment pressure to the lowest possible level. In closed-gas reservoirs, it is not uncommon to recover as much as 90% of the initial gas-in-place.

Cole (1969) points out that for water-drive gas reservoirs, recovery may be rate dependent. There are two possible influences that producing rate may have on ultimate recovery. First, in an active water-drive reservoir, the abandonment pressure may be quite high, sometimes only a few psi below initial pressure. In such a case, the number of pounds of gas remaining in the pore spaces at abandonment will be relatively great.

The encroaching water, however, reduces the initial gas saturation. Therefore, the high abandonment pressure is somewhat offset by the reduction in initial gas saturation. If the reservoir can be produced at a rate greater than the rate of water influx rate, without water coning, then a high producing rate could result in maximum recovery by taking advantage of a combination of reduced abandonment pressure and reduction in initial gas saturation. Second, the water coning problems may be very severe in gas reservoirs, in which case it will be necessary to restrict withdrawal rates to reduce the magnitude of this problem.

Cole suggests that the recovery from water-drive gas reservoirs is substantially less than recovery from closed-gas reservoirs. As a rule of thumb, recovery from a water-drive reservoir will be approximately 50% to 80% of the initial gas-in-place. The structural location of producing wells and the degree of water coning are important considerations in determining ultimate recovery.

A set of circumstances could exist—such as the location of wells very high on the structure with very little coning tendencies—where water-drive recovery would be greater than depletion-drive recovery. Abandonment pressure is a major factor in determining recovery efficiency, and permeability is usually the most important factor in determining the magnitude of the abandonment pressure. Reservoirs with low permeability will have higher abandonment pressures than reservoirs with high permeability. A certain minimum flow rate must be sustained, and a higher permeability will permit this minimum flow rate at a lower pressure.

Tight Gas Reservoirs

Gas reservoirs with permeabilities of less than 0.1 md are considered "tight gas" reservoirs. They present unique problems to reservoir engineers when applying the MBE to predict the gas-in-place and recovery performance.

The use of the conventional material balance in terms of the p/Z plot is a powerful tool for evaluating the performance of gas reservoirs. For a volumetric gas reservoir, the MBE is expressed in different forms that will produce a linear relationship between p/Z versus the cumulative gas production, G_p. Two such forms are given by Equation 13-11:

$$\frac{p}{Z} = \frac{p_i}{Z_i} - \left[\left(\frac{p_i}{Z_i}\right)\frac{1}{G}\right]G_p$$

Simplified:

$$\frac{p}{Z} = \frac{p_i}{Z_i}\left[1 - \frac{G_P}{G}\right]$$

The MBE as expressed by either of these equations is very simple to apply because it is not dependent on flow rates, reservoir configuration, rock properties, or well details. However, there are fundamental assumptions that must be satisfied when applying the equation, including the following:

- There is uniform saturation throughout the reservoir at any time.
- There is little or no pressure variation within the reservoir.
- The reservoir can be represented by a single weighted-average pressure at any time.
- The reservoir is represented by a tank, i.e., constant drainage area, of homogeneous properties.

Payne (1996) pointed out that the assumption of uniform pressure distributions is required to ensure that pressure measurements taken at different well locations represent true average reservoir pressures. This assumption implies that the average reservoir pressure to be used in the MBE can be described with one pressure value. In high-permeability reservoirs, small pressure gradients exist away from the wellbore, and the average reservoir pressure estimates can be readily made with short-term shut-in buildups or static pressure surveys.

Unfortunately, the concept of the straight-line p/Z plot as described by the conventional MBE fails to produce this linear behavior when applied to tight gas reservoirs that have not established a constant drainage area. Payne (1996) suggests that the essence of the errors associated with the use of p/Z plots in tight gas reservoirs is that substantial pressure gradients exist within the formation, resulting in a violation of the basic tank assumption. These gradients manifest themselves in terms of scattered, generally curved, and rate-dependent p/Z plot behavior. This nonlinear behavior of p/Z plots, as shown in Figure 13-15, may significantly underestimate gas initially in place (GIIP) when interpreting by the conventional straight-line method. Figure 13-15a reveals that the reservoir pressure declines very rapidly, as the area surrounding the well cannot be recharged as fast as it is depleted by the well. This early, rapid pressure decline is seen often in tight gas reservoirs and is an indication that the use of p/Z plot analysis may be inappropriate. It is quite apparent that the use of early points would dramatically underestimate GIIP, as shown in Figure 13-15a for the Waterton gas field, which has an apparent GIIP of 7.5 Bm^3. However, late time production and pressure data show a nearly double GIIP of 16.5 Bm^3, as shown in Figure 13-15b.

The main problem with tight gas reservoirs is the difficulty of accurately estimating the average reservoir pressure required for p/Z plots as a function of G_p or time. If the pressures obtained during shut-in do not

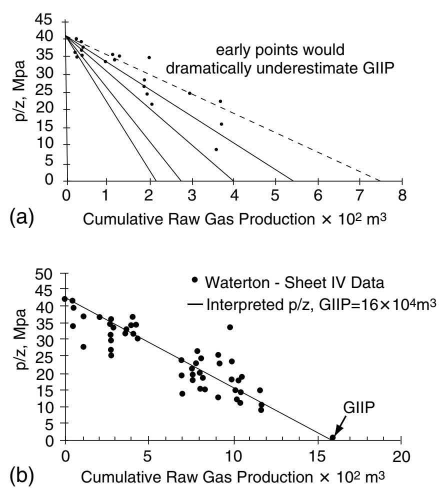

Figure 13-15. (a) Real-life example of p/Z plot from Sheet IVc in the Waterton Gas Field. **(b)** Real-life example of p/Z plot from Sheet IV in the Waterton Gas Field.

reflect the average reservoir pressure, the resulting analysis will be inaccurate. In tight gas reservoirs, excessive shut-in times of months or years may be required to obtain accurate estimates of average reservoir pressure. The minimum shut-in time required to obtain a reservoir pressure that represents the average reservoir pressure must be at least equal to the time it takes to reach the pseudosteady-state, t_{pss}. This time, for a well in the center of a circular or square drainage area, is given by:

$$t_{pss} = \frac{15.8\phi\mu_{gi}c_{ti}A}{k}$$

with

$$c_{ti} = S_{wi}\, c_{wi} + S_g\, c_{gi} + c_f$$

where t_{pss} = stabilization (pseudosteady-state) time, days
c_{ti} = total compressibility coefficient at initial pressure, psi^{-1}
c_{wi} = water compressibility coefficient at initial pressure, psi^{-1}
c_f = formation compressibility coefficient, psi^{-1}
c_{gi} = gas compressibility coefficient at initial pressure, psi^{-1}
φ = porosity, fraction

Since most tight gas reservoirs are hydraulically fractured, Earlougher (1977) proposed the following expression for estimating the minimum shut-in time to reach the semisteady-state:

$$t_{p_{ss}} = \frac{474\phi\mu_g c_t x_f^2}{k}$$

where x_f = fracture half-length, ft
k = permeability, md

Example 13-5

Estimate the time required for a shut-in gas well to reach its 40-acre drainage area. The well is located in the center of a square-drainage boundary with the following properties:

$\phi = 14\%$
$\mu_{gi} = 0.016$ cp
$c_{ti} = 0.0008$ psi^{-1}
$A = 40$ acres
$K = 0.1$ md

Solution

Calculate the stabilization time by applying Earlougher's equation to give:

$$t_{pss} = \frac{474\phi\mu_g c_i x_f^2}{k}$$

$$t_{pss} = \frac{15.8\ (0.14)\ (0.016)\ (0.0008)\ (40)\ (43{,}650)}{0.1} = 493 \text{ days}$$

This example indicates that an excessive shut-in time of approximately 16 months is required to obtain a reliable average reservoir pressure.

Unlike curvature in the p/Z plot, which can be caused by

- An aquifer
- An oil leg
- Formation compressibility, or
- Liquid condensation

scatter in the p/Z plot is diagnostic of substantial reservoir pressure gradients. Hence, if substantial scatter is seen in a p/Z plot, the tank assumption is being violated and the plot should not be used to determine the GIIP. One obvious solution to the material balance problem in tight gas reservoirs is the use of a numerical simulator. Two other relatively new approaches to solving the material balance problem that can be used if reservoir simulation software is not available are:

- The compartmental reservoir approach
- The combined decline-curve and type-curve approach

These two methodologies are discussed next.

Compartmental Reservoir Approach

A compartmental reservoir is defined as a reservoir that consists of two or more distinct regions that are allowed to communicate. Each compartment or "tank" is described by its own material balance, which is coupled to the material balance of the neighboring compartments through influx or efflux gas across the common boundaries. Payne (1996) and Hagoort and Hoogstra (1999) proposed two different robust and rigorous schemes for the numerical solution of the material balance equations of compartmental gas reservoirs. The main difference between the two approaches is that Payne solves for the pressure in each compartment **explicitly** and Hagoort and Hoogstra do so **implicitly**. However, *both schemes* employ the following basic approach:

- Divide the reservoir into a number of compartments with each compartment containing one or more production wells that are proximate and that measure consistent reservoir pressures. The initial division should be made with as few tanks as possible, and each compartment should have different dimensions in terms of length L, width W, and height h.
- Each compartment must be characterized by a historical production and pressure decline data as a function of time.
- If the initial division is not capable of matching the observed pressure decline, additional compartments can be added either by subdividing the previously defined tanks or by adding tanks that do not contain drainage points, that is, production wells.

The practical application of the compartmental reservoir approach is illustrated by the following two methods:

- Payne's method
- Hagoort–Hoogstra method

Payne's Method

Rather than using the conventional single-tank MBE in describing the performance of tight gas reservoirs, Payne (1996) suggests a different approach that is based on subdividing the reservoir into a number of tanks, that is, compartments, which are allowed to communicate. Such compartments can either be depleted directly by wells or indirectly through other tanks. Flow rate between tanks is set proportionally to either the difference in the squares of tank pressure or the difference in pseudo-pressures, m(p). To illustrate the concept, consider a reservoir that consists of two compartments, 1 and 2, as shown schematically in Figure 13-16.

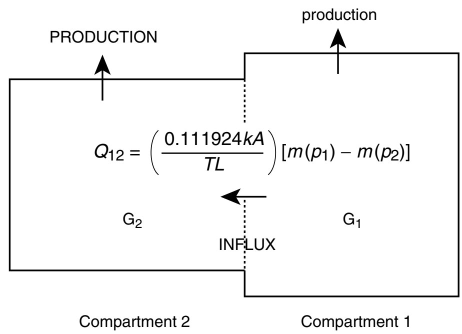

Figure 13-16. Schematic representation of compartmental reservoir consisting of two reservoir compartments separated by a permeable boundary.

Initially, that is, before the start of production, the two compartments are in equilibrium, with the same initial reservoir pressure. Gas production can be produced from either one or both compartments. With gas production, the pressures in the reservoir compartments will decline at different rates depending on the production rate from each compartment and the crossflow rate between the two compartments. Adopting the convention that influx is positive if gas flows from compartment 1 into compartment 2, the linear gas flow rate between the two compartments in terms of gas pseudo-pressure is given by Equation 6-23 from Chapter 6:

$$Q_{12} = \left(\frac{0.111924\,k\,A}{T\,L}\right)\left[m(p_1) - m(p_2)\right]$$

where Q_{12} = flow rate between the two compartments, scf/day
$m(p_1)$ = gas pseudo-pressure in compartment (tank) 1, psi^2/cp
$m(p_2)$ = gas pseudo-pressure in compartment (tank) 2, psi^2/cp
k = permeability, md
L = distance between the center of the two compartments, ft
A = cross-sectional area, width × height, ft^2
T = temperature, °R

This equation can be expressed in a more compacted form by including a "communication factor," C_{12}, between the two compartments:

$$Q_{12} = C_{12}\left[m(p_1) - m(p_2)\right] \qquad (13\text{-}34)$$

The C_{12} between the two compartments is computed by calculating the individual communication factor for each compartment and employing an averaging technique. The communication factor for each of the compartments is as follows:

For compartment 1: $C_1 = \dfrac{0.111924 k_1 \; A_1}{T \; L_1}$

For compartment 2: $C_2 = \dfrac{0.111924 k_2 \; A_2}{T \; L_2}$

And the communication factor between the two compartments, C_{12}, is given by the following harmonic average technique:

$$C_{12} = \frac{2 C_1 \; C_2}{(C_1 + C_2)}$$

where

C_{12} = communication factor between two compartments, scf/day/psi^2/cp
C_1 = communication factor for compartment 1, scf/day/psi^2/cp
C_2 = communication factor for compartment 2, scf/day/psi^2/cp
L_1 = length of compartment 1, ft
L_2 = length of compartment 2, ft
A_1 = cross-sectional area of compartment 1, ft^2
A_2 = cross-sectional area of compartment 2, ft^2

The cumulative gas influx, G_{p12}, from compartment 1 to compartment 2 is given by the integration of flow rate over time t as:

$$G_{p12} = \int_0^t Q_{12} \; dt = \sum_0^t (\Delta Q_{12}) \Delta t \tag{13-35}$$

Payne proposes that **individual** compartment pressures are determined by assuming a straight-line relationship of p/Z versus G_{pt}, with the total gas production, G_{pt}, from an individual compartment as defined by the following expression:

$G_{pt} = G_p + G_{p12}$

where G_p is cumulative gas produced from wells in the compartment and G_{p12} is the cumulative gas efflux/influx between the connected compartments. Solving Equation 8-59 for the pressure in each compartment and assuming a positive flow from compartment 1 to compartment 2 gives:

$$p_1 = \left(\frac{p_i}{Z_i}\right) Z_1 \left[1 - \frac{G_{p1} + G_{p12}}{G_1}\right] \tag{13-36}$$

$$p_2 = \left(\frac{p_i}{Z_i}\right) Z_2 \left[1 - \frac{G_{p2} - G_{p12}}{G_2}\right] \tag{13-37}$$

with:

$$G_1 = 43{,}560\, A_1\, h_1\, \phi_1\, (1 - S_{wi}) / B_{gi} \tag{13-38}$$
$$G_2 = 43{,}560\, A_2\, h_2\, \phi_2\, (1 - S_{wi}) / B_{gi} \tag{13-39}$$

where G_1 = initial gas-in-place in compartment 1, scf
G_2 = initial gas-in-place in compartment 2, scf
G_{p1} = **actual** cumulative gas production from compartment 1, scf
G_{p2} = **actual** cumulative gas production from compartment 2, scf
A_1 = areal extent of compartment 1, acres
A_2 = areal extent of compartment 2, acres
h_1 = average thickness of compartment 1, ft
h_2 = average thickness of compartment 2, ft
B_{gi} = initial gas formation volume factor, ft^3/scf
ϕ_1 = average porosity in compartment 1
ϕ_2 = average porosity in compartment 2

The subscripts 1 and 2 denote the two compartments 1 and 2, while the subscript i refers to an initial condition. The required input data for Payne's method are as follows:

- Amount of gas contained in each tank, that is, tank dimensions, porosity, and saturation
- Inter-compartment communication factors, C_{12}
- Initial pressure in each compartment
- Production data profiles from the individual tanks

Payne's technique is performed fully explicit in time. At each time step, the pressures in various tanks are calculated, yielding a pressure profile that can be matched to the actual pressure decline. The specific steps of this iterative method are summarized below:

Step 1. Prepare the available gas properties data in tabulated and graphical forms including:
-Z versus p
-μ_g versus p
-$2p/(\mu_g Z)$ versus p
-m(p) versus p

Step 2. Divide the reservoir into compartments and determine the dimensions of each compartments in terms of:
-Length, L
-Height, h
-Width, W
-Cross-sectional area, A

Step 3. For each compartment, determine the initial gas-in-place, G. Assuming two compartments, for example, then calculate G_1 and G_2 from Equations 13-38 and 13-39.

$$G_1 = 43{,}560\, A_1\, h_1\, \phi_1\, (1-S_{wi}) / B_{gi}$$
$$G_2 = 43{,}560\, A_2\, h_2\, \phi_2\, (1-S_{wi}) / B_{gi}$$

Step 4. For each compartment, make a plot of p/Z vs. G_P that can be constructed by simply drawing a straight line between p_i/Z_i with initial gas-in-place in both compartments, G_1 and G_2.

Step 5. Calculate the communication factors for each compartment and between compartments. For two compartments:

$$C_1 = \frac{0.111924\, k_1\, A_1}{T\, L_1}$$

$$C_2 = \frac{0.111924\, k_2\, A_2}{T\, L_2}$$

$$C_{12} = \frac{2\ C_1\ C_2}{(C_1 + C_2)}$$

Step 6. Select a small time step, Δt, and determine the corresponding *actual* cumulative gas production, G_p, from each compartment. Assign $G_p = 0$ if the compartment does not include a well.

Step 7. Assume (guess) the pressure distributions throughout the selected compartmental system and determine the gas deviation factor, Z, at each pressure. For a two-compartment system, let the initial values be denoted by p_1^k and p_2^k.

Step 8. Using the assumed values of pressure, p_1^k and p_2^k, determine the corresponding $m(p_1)$ and $m(p_2)$ from the data of Step 1.

Step 9. Calculate the gas influx rate, Q_{12}, and cumulative gas influx G_{p12} by applying Equations 13-34 and 13-35, respectively.

$$Q_{12} = C_{12}\ [m(p_1) - m(p_2)]$$

$$G_{p12} = \int_0^t Q_{12}\ dt = \sum_0^t (\Delta Q_{12})\,\Delta t$$

Step 10. Substitute the values of G_{p12}, Z-factor, and actual values of G_{P1} and G_{P2}, in Equations 13-36 and 13-37 to calculate the pressure in each compartment, denoted by p_i^{k+1} and p_2^{k+1}.

$$p_1^{k+1} = \left(\frac{p_i}{Z_i}\right) Z_1 \left(1 - \frac{G_{p1} + G_{p12}}{G_1}\right)$$

$$p_2^{k+1} = \left(\frac{p_i}{Z_i}\right) Z_2 \left(1 - \frac{G_{p2} - G_{p12}}{G_2}\right)$$

Step 11. Compare the assumed and calculated values, $|p_1^k - p_i^{k+1}|$ and $|p_2^k - p_2^{k+1}|$. If a satisfactory match is achieved within a tolerance of 5–10 psi for all the pressure values, then Steps 3 through 7 are repeated at a new time level with corresponding historical gas production data. If the match is not satisfactory, repeat the iterative cycle of Steps 4 through 7 and set $p_1^k = p_i^{k+1}$ and $p_2^k = p_2^{k+1}$.

Step 12. Repeat Steps 6 through 11 to produce a pressure-decline profile for each compartment that can be compared with the actual pressure profile for each compartment or that from Step 4.

Performing a material-balance history match consists of varying the number of compartments required, the dimension of the compartments, and the communication factors until an acceptable match of the pressure decline is obtained. The improved accuracy in estimating the original gas-in-place, resulting from determining the optimum number and size of compartments, stems from the ability of the proposed method to incorporate reservoir pressure gradients, which are completely neglected in the single-tank conventional p/Z plot method.

Hagoort–Hoogstra Method

Based on Payne's method, Hagoort and Hoogstra (1999) developed a numerical method to solve the MBE of compartmental gas reservoirs that employs an implicit, iterative procedure, and that recognizes the pressure dependency of the gas properties. The iterative technique relies on adjusting the size of the compartments and the transmissibility values to match the historical pressure data for each compartment as a function of time. Referring to Figure 13-16, the authors assume a thin permeable layer with a transmissibility of Γ_{12} separating the two compartments. Hagoort and Hoogstra expressed the instantaneous gas influx through the thin permeable layer by Darcy's equation, as given by (in Darcy's units):

$$Q_{12} = \frac{\Gamma_{12}\,(p_1^2 - p_2^2)}{2\,p_1\,(\mu_g\,B_g)_{avg}}$$

where Γ_{12} = the transmissibility between compartments

The gas influx between compartments can be obtained by modifying Equation 6-23 in Chapter 6 to give:

$$Q_{12} = \frac{0.111924\,\Gamma_{12}\,(p_1^2 - p_2^2)}{T\,L} \tag{13-40}$$

with

$$\Gamma_{12} = \frac{\Gamma_1 \Gamma_2 \, (L_1 + L_2)}{L_1 \Gamma_2 + L_2 \Gamma_1} \tag{13-41}$$

$$\Gamma_1 = \left[\frac{kA}{Z\mu_g} \right]_1 \tag{13-42}$$

$$\Gamma_2 = \left[\frac{kA}{Z\mu_g} \right]_2 \tag{13-43}$$

where Q_{12} = influx gas rate, scf/day
L = distance between the centers of compartment 1 and 2, ft
A = cross-sectional area, ft^2
μ_g = gas viscosity, cp
Z = gas deviation factor
k = permeability, md
p = pressure, psia
T = temperature, °R
L_1 = length of compartment 1, ft
L_2 = length of compartment 2, ft

The subscripts 1 and 2 refer to compartments 1 and 2, respectively.

The material balance for the two reservoir compartments can be modified to include the gas influx from compartment 1 to compartment 2:

$$\frac{p_1}{Z_1} = \frac{p_1}{Z_1} \left(1 - \frac{G_{p1} + G_{p12}}{G_1} \right) \tag{13-44}$$

$$\frac{p_2}{Z_2} = \frac{p_1}{Z_1} \left(1 - \frac{G_{p2} - G_{p12}}{G_2} \right) \tag{13-45}$$

where p_1 = initial reservoir pressure, psi
Z_1 = initial gas deviation factor
G_p = actual (historical) cumulative gas production, scf
G_1, G_2 = initial gas-in-place in compartments 1 and 2, scf
G_{p12} = cumulative gas influx from compartment 1 to 2, scf as given in Equation 13-35

Again, subscripts 1 and 2 represent compartments 1 and 2, respectively.

To solve the material balance equations as represented by the relationships in Equations 13-45 and 13-46 for the two unknowns p_1 and p_2, the two expressions can be arranged to equate to zero, as follows:

$$F_1\ (p_1, p_2) = p_1 - \left(\frac{p_i}{Z_i}\right) Z_1 \left(1 - \frac{G_{p1} + G_{p12}}{G_1}\right) = 0 \tag{13-46}$$

$$F_2\ (p_1, p_2) = p_2 - \left(\frac{p_i}{Z_i}\right) Z_2 \left(1 - \frac{G_{p2} - G_{p12}}{G_2}\right) = 0 \tag{13-47}$$

The general methodology of applying the method is very similar to that of Payne's and involves the following specific steps:

Step 1. Prepare the available data on gas properties in tabulated and graphical forms that include Z versus p and μ_g versus p

Step 2. Divide the reservoir into compartments and determine the dimensions of each compartment in terms of

- Length, L
- Height, h
- Width, W
- Cross-sectional area, A

Step 3. For each compartment, determine the initial gas-in-place, G. For clarity, assume two gas compartments and calculate G_1 and G_2 from Equations 13-38 and 13-39.

$$G_1 = 43{,}560\ A_1\ h_1\ \phi_1\ (1 - S_{wi})\ /\ B_{gi}$$
$$G_2 = 43{,}560\ A_2\ h_2\ \phi_2\ (1 - S_{wi})\ /\ B_{gi}$$

Step 4. For each compartment, make a plot of p/Z versus G_P that can be constructed by simply drawing a straight line between p_i/Z_i with initial gas-in-place in both compartments, G_1 and G_2.

Step 5. Calculate the transmissibility by applying Equation 13-41.

Step 6. Select a time step, Δt,and determine the corresponding actual cumulative gas production G_{p1} and G_{p2}.

Step 7. Calculate the gas influx rate, Q_{12}, and cumulative gas influx, G_{p12}, by applying Equations 13-40 and 13-35, respectively.

$$Q_{12} = \frac{0.111924\ \Gamma_{12}\,(p_1^2 - p_2^2)}{T\,L}$$

$$G_{p12} = \int_0^t Q_{12}\ dt = \sum_0^t (\Delta Q_{12})\,\Delta t$$

Step 8. Start the iterative solution by assuming initial estimates of the pressure for compartments 1 and 2 (i.e., p_1^k and p_2^k). Using Newton and Raphson's iterative scheme, calculate new improved values of the pressure p_1^{k+1} and p_2^{k+1} by solving the following linear equations as expressed in a matrix form:

$$\begin{bmatrix} p_1^{k+1} \\ \\ p_2^{k+1} \end{bmatrix} = \begin{bmatrix} p_1^k \\ \\ p_2^k \end{bmatrix} - \begin{bmatrix} \dfrac{\partial F_1(p_1^k, p_2^k)}{\partial p_1} & \dfrac{\partial F_1(p_1^k, p_2^k)}{\partial p_2} \\ \dfrac{\partial F_2(p_1^k, p_2^k)}{\partial p_1} & \dfrac{\partial F_2(p_1^k, p_2^k)}{\partial p_2} \end{bmatrix}^{-1} \begin{bmatrix} -F_1(p_1^k, p_2^k) \\ \\ -F_2(p_1^k, p_2^k) \end{bmatrix}$$

where the superscript "−1" denotes the inverse of the matrix. The partial derivatives in this system of equations can be expressed in analytical form by differentiating Equations 8-140 and 8-141 with respect to p_1 and p_2. During an iterative cycle, the derivatives are evaluated at the updated new pressures, i.e., p_1^{k+1} and p_2^{k+1}. The iteration is stopped when $|p_1^{k+1} - p_i^k|$ and $|p_2^{k+1} - p_2^k|$ are less than a certain pressure tolerance, that is, 5–10 psi.

Step 9. Generate the pressure profile as a function of time for each compartment by repeating Steps 2 and 3.

$$\frac{p_1}{Z_1} = \frac{p_1}{Z_1}\left(1 - \frac{G_{p1} + G_{p12}}{G_1}\right)$$

where p_1 = initial reservoir pressure, psi
Z_1 = initial gas deviation factor
G_p = actual (historical) cumulative gas production, scf
G_1, G_2 = initial gas-in-place in compartment 1 and 2, scf
G_{p12} = cumulative gas influx from compartment 1 to 2 in scf, as given in Equation 13-35

Step 10. Repeat Steps 6 through 11 to produce a pressure decline profile for each compartment that can be compared with the actual pressure profile for each compartment or that from step 4.

Compare the calculated pressure profiles with those of the observed pressures. If a match has not been achieved, adjust the size and number of compartments (i.e., initial gas-in-place) and repeat Steps 2 through 10.

Shallow Gas Reservoirs

Tight shallow gas reservoirs present a number of unique challenges in determining reserves accurately. Traditional methods such as decline analysis and material balance are inaccurate owing to the formation's low permeability and the usually poor-quality pressure data. The low permeabilities cause long transient periods that are not separated early from production decline with conventional decline analysis, resulting in lower confidence in selecting the appropriate decline characteristics, which affects recovery factors and remaining reserves significantly. In an excellent paper by West and Cochrane (1995), the authors used the Medicine Hat field in Western Canada as an example of these types of reservoirs and developed a methodology called the Extended Material Balance (EMB) technique, to evaluate gas reserves and potential infill drilling.

The Medicine Hat field is a tight, shallow gas reservoir producing from multiple highly interbedded, silty sand formations with poor permeabilities of < 0.1 md. This poor permeability is the main characteristic of these reservoirs that affects conventional decline analysis. Owing to these low permeabilities, and in part to commingled multilayer production effects, wells experience long transient periods before they begin experiencing the pseudosteady-state flow that represents the decline portion of their lives. One of the principal assumptions often neglected when conducting decline analysis is that pseudosteady-state must have been achieved. The initial transient production trend of a well or group of wells is not indicative of the long-term decline of the well. Distinguishing the transient production of a well from its pseudosteady-state production is often difficult, and this can lead to errors in determining the decline characteristic (exponential, hyperbolic, or harmonic) of a well. Figure 13-17 shows the production history from a tight, shallow gas well and illustrates the difficulty in selecting the correct decline. Another characteristic of tight, shallow gas reservoirs that affects conventional decline analysis is that constant reservoir conditions, an assumption required for conventional decline analysis, do not exist because of increasing drawdown, changing operating strategies, erratic development, and deregulation.

Material balance is affected by tight, shallow gas reservoirs because the pressure data are limited, of poor quality, and not representative of a majority of the wells. Because the risk of drilling dry holes is low and drillstem tests (DSTs) are not cost-effective in the development of shallow gas, DST data are very limited. Reservoir pressures are recorded only for government-designated "control" wells, which account for only 5% of all wells. Shallow gas produces from multiple formations, and production from these formations is typically commingled, exhibiting some degree of pressure equalization. Unfortunately, the control wells are segregated by tubing/packers, and consequently, the control-well pressure data are not representative of most commingled wells. In addition, pressure monitoring has been very inconsistent. Varied measurement points (downhole or

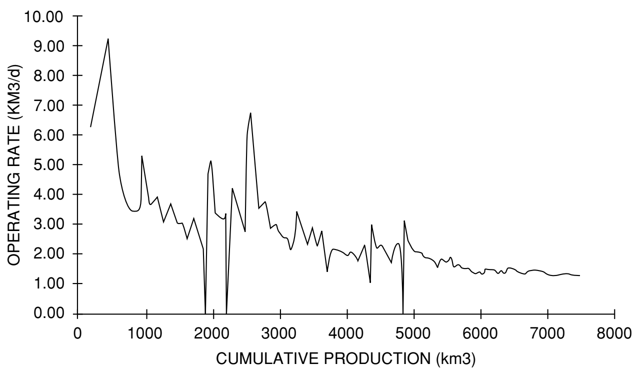

Figure 13-17. Production history for a typical Medicine Hat property (Permission to copy SPE, copyright SPE 1995).

wellhead), inconsistent shut-in times, and different analysis types (e.g., buildup and static gradient) make quantitative pressure-tracking difficult. As Figure 13-18 shows, both of these problems result in a scatter of data, which makes material balance extremely difficult.

Wells in the Medicine Hat shallow gas area are generally cased, perforated, and fractured in one, two, or all three formations, as ownerships vary not only areally but between formations. The Milk River and Medicine Hat formations are usually produced commingled. Historically, the Second White Specks formation has been segregated from the other two; recently, however, commingled production from all three formations has been approved. Spacing for shallow gas is usually two to four wells per section. As a result of the poor reservoir quality and low pressure, well productivity is very low. Initial rates rarely exceed 700 Mscf/day. Current average production per well is approximately 50 Mscf/day for a three-formation completion. There are approximately 24,000 wells producing from the Milk River formation in Southern Alberta and Saskatchewan with total estimated gas reserves of 5.3 Tscf. West and Cochrane's EMB technique was developed to determine gas reserves in 2,300 wells in the Medicine Hat field.

The EMB technique is essentially an iterative process for obtaining a suitable p/Z versus G_p line for a reservoir where pressure data are inadequate. It combines the principles of volumetric gas depletion with the gas-deliverability (back-pressure) equation. The deliverability equation for radial flow of gas describes the relationship between the pressure differential in the wellbore and the gas flow rate from the well:

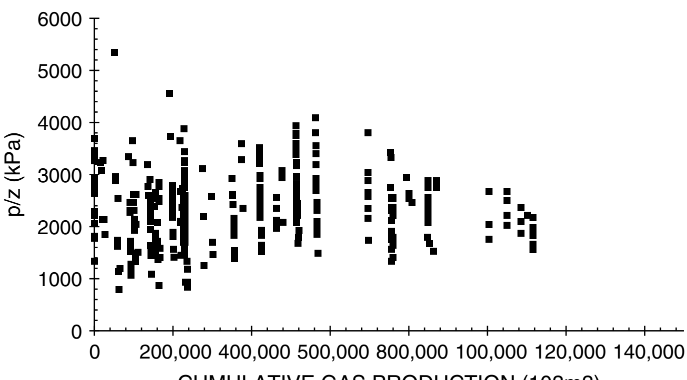

Figure 13-18. Scatter pressure data for a typical Medicine Hat property (Permission to copy SPE, copyright SPE 1995).

$$Q_g = C\,[p_r^2 - p_{wf}^2]^n$$

Owing to the very low production rates from the wells in Medicine Hat shallow gas, a laminar flow regime exists, which can be described with an exponent n = 1. The terms making up the coefficient C in the back-pressure equation are either fixed reservoir parameters (kh, r_e, r_w, and T) that do not vary with time or terms that fluctuate with pressure, temperature, and gas composition, for example, μ_g and Z. The performance coefficient "C" is given by:

$$C = \frac{k\,h}{1422\,T\,\mu_g Z[\ln(r_e / r_w) - 0.5]}$$

Because the original reservoir pressure in these shallow formations is low, the differences between initial and abandonment pressures are not significant and the variation in the pressure-dependent terms over time can be assumed negligible. C may be considered constant for a given Medicine Hat shallow gas reservoir over its life. With these simplifications for shallow gas, the deliverability equation becomes:

$$Q_g = C\,[p_r^2 - p_{wf}^2]$$

The sum of the instantaneous production rates with time will yield the relationship between G_p and reservoir pressure, similar to the material balance equation. By use of this common relationship, with the unknowns being reservoir pressure, p, and the performance coefficient, C, the EMB method involves iterating to find the correct p/Z versus G_p relationship to give a constant C with time. The proposed iterative method is applied as outlined in the following steps:

Step 1. To avoid calculating individual reserves for each of the 2,300 wells, West and Cochrane grouped wells by formation and by date on production. The authors verified this simplification on a test group by ensuring that the reserves from the group of wells yielded the same results as the sum of the individual well reserves. These groupings were used for each of the 10 properties, and the results of the groupings combined to give a property production forecast. Also, to estimate the reservoir decline characteristics more accurately, the rates were normalized to reflect changes in the bottom-hole flowing pressure (BHFP).

Step 2. Using the gas specific gravity and reservoir temperature, calculate the gas deviation factor, Z, as a function of pressure and plot p/Z versus p on a Cartesian scale.

Step 3. An initial estimate for the p/Z variation with G_p is made by guessing an initial pressure, p_i, and a linear slope, m, of Equation 13-11:

$$\frac{p}{Z} = \frac{p_i}{Z_i} - [m] G_p$$

with the slope m defined by:

$$m = \left(\frac{p_i}{Z_i}\right)\frac{1}{G}$$

Step 4. Starting at the initial production date for the property, the p/Z versus time relationship is established by simply substituting the actual cumulative production, G_p, into the MBE with estimated slope m and p_i because actual cumulative production G_p versus time is known. The reservoir pressure, p, can then be constructed as a function of time from the plot of p/Z as a function of p, that is, Step 2.

Step 5. Knowing the actual production rates, Q_g, and bottom-hole flowing pressures, p_{wf}, for each monthly time interval and having estimated reservoir pressures, p, from Step 3, C is calculated for each time interval with:

$$C = \frac{Q_g}{p^2 - p_{wf}^2}$$

Step 6. C is plotted against time. If C is not constant (i.e., the plot is not a horizontal line), a new p/Z versus G_p is guessed and the process is repeated from Step 3 through Step 5.

Step 7. Once a constant C solution is obtained, the representative p/Z relationship has been defined for reserves determination.

Use of the EMB method in the Medicine Hat shallow gas makes the fundamental assumptions (1) that the gas pool depletes volumetrically (i.e., there is no water influx) and (2) that all wells behave like an average well with the same deliverability constant, turbulence constant, and BHFP, which is a reasonable assumption given the number of wells in the area, the homogeneity of the rocks, and the observed well production trends.

In the EMB evaluation, West and Cochrane point out that wells for each property were grouped according to their producing interval so that the actual production from the wells could be related to a particular reservoir pressure trend. When calculating the coefficient C, as outlined above, a total C based on grouped production was calculated and then divided by the number of wells producing in a given time interval to give an average C value. The average C value was used to calculate an average permeability/thickness, kh, for comparison with actual kh data obtained through buildup analysis for the reservoir, as follows:

$$kh = 1422\,T\mu_g Z[\ln(r_e/r_w) - 0.5]C$$

For that reason, kh versus time, instead of C versus time, was plotted in the method. Figure 13-19 shows a flat kh versus time profile indicating a valid p/Z versus G_p relationship.

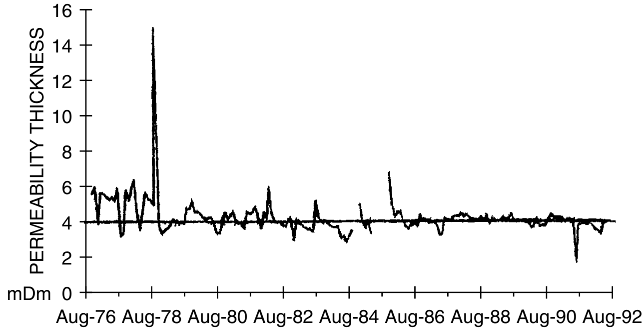

Figure 13-19. Example for a successful EMB solution-flat kh profile (Permission to copy SPE, copyright SPE 1995).

PROBLEMS

1. The following information is available on a volumetric gas reservoir:

   Initial reservoir temperature, T_i = 155°F
   Initial reservoir pressure, p_i = 3500 psia
   Specific gravity of gas, γ_g = 0.65 (air = 1)
   Thickness of reservoir, h = 20 ft
   Porosity of the reservoir, ϕ = 10%
   Initial water saturation, S_{wi} = 25%

   After producing 300 MMscf, the reservoir pressure declined to 2,500 psia. Estimate the areal extent of this reservoir.

2. The following pressures and cumulative production data[2] are available for a natural gas reservoir:

Reservoir Pressure, psia	Gas Deviation Factor, z	Cumulative Production, MMMscf
2080	0.759	0
1885	0.767	6.873
1620	0.787	14.002
1205	0.828	23.687
888	0.866	31.009
645	0.900	36.207

   a. Estimate the initial gas-in-place.
   b. Estimate the recoverable reserves at an abandonment pressure of 500 psia. Assume z_a = 1.00.
   c. What is the recovery factor at the abandonment pressure of 500 psia?

3. A gas field with an active water drive showed a pressure decline from 3,000 to 2,000 psia over a 10-month period. From the following production data, match the past history and calculate the original hydrocarbon gas in the reservoir. Assume z = 0.8 in the range of reservoir pressures and T = 140°F.

Data					
t, months	0	2.5	5.0	7.5	10.0
p, psia	3000	2750	2500	2250	2000
G_p, MMscf	0	97.6	218.9	355.4	500.0

[2]Ikoku, C., *Natural Gas Reservoir Engineering,* John Wiley and Sons, 1984.

4. A volumetric gas reservoir produced 600 MMscf of 0.62 specific gravity gas when the reservoir pressure declined from 3,600 to 2,600 psi. The reservoir temperature is reported at 140°F. Calculate:

   a. Gas initially in place
   b. Remaining reserves to an abandonment pressure of 500 psi
   c. Ultimate gas recovery at abandonment

5. The following information on a water-drive gas reservoir is given:

   Bulk volume = 100,000 acre-ft
   Gas gravity = 0.6
   Porosity = 15%
   S_{wi} = 25%
   T = 140°F
   p_i = 3500 psi

   Reservoir pressure has declined to 3,000 psi while producing 30 MMMscf of gas and no water production. Calculate cumulative water influx.

6. The pertinent data for the Mobil-David field are given below.

G = 70 MMMscf	p_i = 9507 psi	ϕ = 24%	S_{wi} = 35%
$c_w = 401 \times 10^{-6}$ psi^{-1}	$c_f = 3.4 \times 10^{-6}$ psi^{-1}	γ_g = 0.94	T = 266°F

   For this volumetric abnormally pressured reservoir, calculate and plot cumulative gas production as a function of pressure.

7. The Big Butte field is a volumetric dry-gas reservoir with a recorded initial pressure of 3,500 psi at 140°F. The specific gravity of the produced gas is measured at 0.65. The following reservoir data are available from logs and core analysis:

   Reservoir area = 1500 acres
   Thickness = 25 ft
   Porosity = 15%
   Initial water saturation = 20%

   Calculate:

   a. Initial gas in place as expressed in scf
   b. Gas viscosity at 3,500 psi and 140°F

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